

Homework 1 CPSC 346

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1. The following tables are the current status of the registers and stack memory for a 64-bit computer. Please explain how the following instructions change the registers and the stack. Assume a) and b) operate on the original data as shown in the two tables.

Register	Value
%rax	4
%rbx	8
%rsp	0x00FFEE88

Memory(Stack):

Address	Data
0x00FFEE88	15

a) `push %rax`

Answer:

if you push %rax to the stack, %rsp would point to a location holding 4. and then the stack here would grow downward. specifically, %rsp would go down 8 since its 64 bit, and then %rsp would be pointing to 0x00FFEE80, with the value 4.

b) `pop %rbx`

Answer:

after pop %rbx, becomes 15, and %rsp is 0x00FFEE90, adding 8 now.

2. (10 points) Fill in the value of the registers and the stack memory. Before executing the instruction “`callq _Subroutine,`” we have `%rsp = 0xFF08;` (`%rsp`) = 0x8, which is the “top” value on the stack.

```
_caller:
.
.
400544: callq  _Subroutine
400549: mov    %rax, (%rbx)
.
.
_Subroutine:
400650: mov    %rdi,%rax
.
.
400657: retq
```

1) About to execute the instruction “callq _Subroutine”:

%rip: _____

2) Right after the instruction “callq _Subroutine” is executed:

(Program will move to the subroutine but not start executing the subroutine yet.)

%rip: _____

%rsp: _____

(%rsp): _____

3) About to execute the instruction “retq”:

%rip: _____

%rsp: _____

(%rsp): _____

4) Right after the instruction “retq” is executed:

%rip: _____

%rsp: _____

(%rsp): _____